SIZING AND SELECTING A

Tamil Voice-Cloning Model for On-Device Deployment

Benchmark results, candidate model survey, and runtime evaluation

Lightweight English Cloning Models: Landscape

"Lightweight" = under 200M params AND genuine zero-shot cloning

Model	Params	Architecture	Size	Voice clone?	License
ZipVoice-Distill ★	123M	NAR, flow-matching	~104 MB	**Yes**	Apache-2.0
Chatterbox-Nano	110M	AR + 1-step decoder	Small (FP)	**Yes**	MIT
GPT-SoVITS	~77M	Hybrid AR + VITS	Small	**Yes**	MIT
Chatterbox-Turbo	350M	AR + distilled decoder	Not quantized	**Yes**	MIT
Kokoro-82M (ref.)	82M	NAR (StyleTTS2)	~80–100 MB	**No**	Apache-2.0

Key takeaway:

ZipVoice-Distill is the smallest genuine cloner with a native sherpa-onnx path already validated on all four devices.

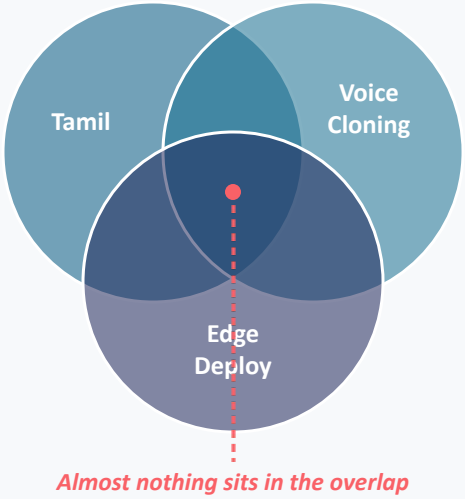
THE PROBLEM

01 / The starting point

An English voice-cloning app already works, built on Flutter + sherpa-onnx. The Tamil version needs a model sized correctly for real devices — not guessed.

02 / The research gap

Tamil support, voice cloning, and edge/on-device deployment rarely co-exist in one model. Most candidates satisfy at most two of the three.



Goal: Use measured device capacity to select a Tamil-capable candidate that will actually run on the production device.

Benchmarking Configuration

MODEL	PARAMETERS	INT8 SIZE	ARCHITECTURE	TRAINING DATA	CAPABILITY
ZipVoice-Distill	**123M**	**~104 MB**	**NAR, flow-matching** (Zipformer)	**Emilia**	**Zero-shot voice** cloning

Tested Devices

Galaxy S23
Snapdragon 8 Gen 2

RAM **8 GB**
NPU **Yes**

Galaxy Tab S9 FE+
Exynos 1380

RAM **8 GB**
NPU **Yes**

Galaxy M53
Dimensity 900

RAM **8 GB**
NPU **Yes**

Target Device

Production Target

Lenovo Tab K10
MediaTek Helio P22T · up to 4 GB RAM · no NPU

Lower spec than all three tested devices

BASELINE

English Model Test: What Was Measured

MODEL	PARAMETERS	INT8 SIZE	ARCHITECTURE	TRAINING DATA	CAPABILITY
ZipVoice-Distill	123M	~104 MB	NAR, flow-matching (Zipformer)	Emilia	Zero-shot voice cloning

Tested on three Android devices via the current sherpa-onnx Flutter app. This result set is the basis for every comparison later in the deck.

Device	Text Length	RTF (median)	Peak RSS (MB)
Galaxy S23	Short	3.826	1033
	Medium	1.520	1287
	Long	1.288	1562
Galaxy Tab S9 FE+	Short	9.279	1021
	Medium	2.918	1491
	Long	2.508	1772
Galaxy M53	Short	8.415	1021
	Medium	2.782	1219
	Long	2.531	1539

READ THE TABLE

RTF = processing time / audio duration. **Below 1.0** means generation keeps pace with playback.

Peak RSS is the highest physical memory used during synthesis — what actually triggers an out-of-memory kill on Android.

What the Results Show

RTF and peak RSS both rise as device tier drops

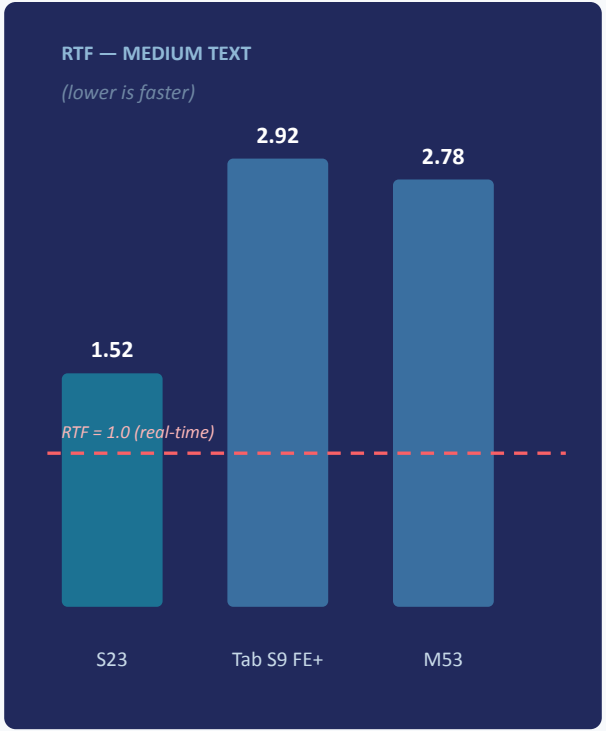
S23 → Tab S9 FE+ → M53 shows a consistent downward trend in headroom, even though all three are 8 GB devices.

Short text has the worst RTF everywhere

3.8× to 9.3× across devices — fixed per-utterance overhead dominates when there's little audio to amortize it over.

RSS already exceeds 1 GB on the highest-spec device

S23, long text: 1562 MB. This is the flagship result — not the constrained tier.



CANDIDATE MODELS — TIER A

Tamil-Native Candidates

IndicF5

AI4BHARAT

PARAMS
~335M

ARCHITECTURE
NAR flow-matching
(F5-TTS backbone)

TAMIL
Native — 1 of 11 Indic languages

TRAINING
1417 hrs

HIGHLIGHTS

Trained on 1417 hours of speech across 11 Indic languages including Tamil. Zero-shot cloning from a reference clip, out of the box.

F5-TTS (base)

NON-INDIC

PARAMS
335.8M

ARCHITECTURE
NAR flow-matching
DiT

TAMIL
Not native — adapter path exists

ADAPTER PROOF
Romanian

HIGHLIGHTS

A lightweight per-language adapter technique has been proven on Romanian — could target Tamil without retraining the full 335M backbone.

Note: *Both are larger than ZipVoice-Distill (123M) — the baseline model that already showed high RTF and RSS on flagship hardware.*

CANDIDATE MODELS — TIER B

Adaptable via Fine-Tuning

XTTS-v2

RULED OUT

PARAMS	~456–467M
ARCHITECTURE	Autoregressive
TAMIL PATH	Proven recipe (used for Hindi, Egyptian Arabic)

Ruled out on size and architecture grounds alone — regardless of how strong its language-adaptation story is.

GPT-SoVITS

WORTH A SPIKE

PARAMS	~77M
ARCHITECTURE	Hybrid AR + VITS
TAMIL PATH	Unofficial — no answer from maintainers yet

Explicit 5-second zero-shot cloning. Already has an ONNX export path. Risk: frozen semantic frontend is Chinese-pretrained.

RUNTIME EVALUATION

LiteRT: Alternative Runtime

Google's actively developed successor to TFLite — framework-agnostic, converting PyTorch, TensorFlow, and JAX models to .tflite.

Snapdragon (S23)

Qualcomm AI Engine Direct

Production-ready

Up to 100× CPU speedup claimed

Dimensity (M53)

MediaTek NeuroPilot

Production-ready

AOT + on-device compilation

Helio P22T (K10)

No delegate available

CPU fallback

Same situation as current ONNX setup

TWO COMPILATION MODES

AOT (Ahead-of-Time)

Best for known target hardware — lower launch memory and init cost.

On-Device / JIT

Best for platform-agnostic distribution of small models.

RUNTIME EVALUATION

LiteRT: Implementation Flow

Conversion path for a PyTorch-trained model (e.g. IndicF5, GPT-SoVITS) — no longer requires going through ONNX as an intermediate step.



CONSTRAINT TO FLAG

The model must be exportable via torch.export — needs checking against whichever Tamil candidate is chosen.

NOT A REPLACEMENT

This would run as a second, parallel pipeline alongside the existing ONNX / sherpa-onnx one — not a swap-out.

SYNTHESIS

Where This Leaves Model Selection

PARAMETER SCALE (LOG)



Only candidate smaller than the tested baseline

Already shows high RTF / RSS on flagship HW

Tamil-native, but larger than a straining baseline

The tension: the smallest model that fits the K10's constraints has no Tamil support, and the only Tamil-native model is larger than a baseline that's already struggling.

CLOSING

Two Candidates Anchor the Next Phase

TAMIL-READY TODAY

IndicF5

- Already speaks Tamil with zero-shot cloning
- Open question: does its 335M NAR footprint fit the K10 at all?
- Needs a LiteRT or ONNX export pipeline built from scratch — not currently exported to either

SIZE-READY TODAY

GPT-SoVITS

- Only cloning candidate smaller than the tested baseline (~77M)
- Existing ONNX export path already in its repo
- Open question: can Tamil be added without degrading cloning quality?

Next steps: test IndicF5 for raw feasibility, spike GPT-SoVITS for Tamil adaptability, and evaluate the LiteRT conversion path for either candidate.